chembl_36 Schema Documentation

ACTION_TYPE:

Table storing the distinct list of action types used in the drug_mechanism table, together with a higher-level parent action type.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ACTION_TYPE	VARCHAR2(50)	NOT NULL	Primary key. Type of action of the drug e.g., agonist, antagonist
	DESCRIPTION	VARCHAR2(200)	NOT NULL	Description of how the action type is used
	PARENT_TYPE	VARCHAR2(50)		Higher-level grouping of action types describing the overall effect on the target e.g., positive vs negative modulator. For example, the AGONIST action type would be classed as POSITIVE MODULATOR, the INHIBITOR action type as NEGATIVE MODULATOR, and the BINDING AGENT action type as OTHER since the overall effect on the target is not specified.

ACTIVITIES:

Activity 'values' or 'end points' that are the results of an assay recorded in a scientific document. Each activity is described by a row.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ACTIVITY_ID	NUMBER	NOT NULL	Unique ID for the activity row
FK	ASSAY_ID	NUMBER	NOT NULL	Foreign key to the assays table (containing the assay description)
FK	DOC_ID	NUMBER		Foreign key to documents table (for quick lookup of publication details - can also link to documents through compound_records or assays table)
FK	RECORD_ID	NUMBER	NOT NULL	Foreign key to the compound_records table (containing information on the compound tested)
FK	MOLREGNO	NUMBER		Foreign key to compounds table (for quick lookup of compound structure - can also link to compounds through compound_records table)
	STANDARD_RELATION	VARCHAR2(50)		Symbol constraining the activity value (e.g. >, <, =)
	STANDARD_VALUE	NUMBER		Same as PUBLISHED_VALUE but transformed to common units: e.g. mM concentrations converted to nM.
	STANDARD_UNITS	VARCHAR2(100)		Selected 'Standard' units for data type: e.g. concentrations are in nM.
	STANDARD_FLAG	NUMBER		Shows whether the standardised columns have been curated/set (1) or just default to the published data (0).
	STANDARD_TYPE	VARCHAR2(250)		Standardised version of the published_activity_type (e.g. IC50 rather than Ic-50/Ic50/ic50/ic-50)
	ACTIVITY_COMMENT	VARCHAR2(4000)		Supplementary information about the measurement. This field should be used to record other information related to the activity data that is supplementary to the overall outcome (e.g. antagonist, poor curve fitting, etc.)
FK	DATA_VALIDITY_COMMENT	VARCHAR2(30)		Comment reflecting whether the values for this activity measurement are likely to be correct - one of 'Manually validated' (checked original paper and value is correct), 'Potential author error' (value looks incorrect but is as reported in the original paper), 'Outside typical range' (value seems too high/low to be correct e.g., negative IC50 value), 'Non standard unit type' (units look incorrect for this activity type).
	POTENTIAL_DUPLICATE	NUMBER		When set to 1, indicates that the value is likely to be a repeat citation of a value reported in a previous ChEMBL paper, rather than a new, independent measurement. Note: value of zero does not guarantee that the measurement is novel/independent though
	PCHEMBL_VALUE	NUMBER		Negative log of selected concentration-response activity values (IC50/EC50/XC50/AC50/Ki/Kd/Potency)
FK	BAO_ENDPOINT	VARCHAR2(11)		ID for the corresponding result type in BioAssay Ontology (based on standard_type)
	UO_UNITS	VARCHAR2(10)		ID for the corresponding unit in Unit Ontology (based on standard_units)
	QUDT_UNITS	VARCHAR2(70)		ID for the corresponding unit in QUDT Ontology (based on standard_units)
	TOID	INTEGER		The Test Occasion Identifier, used to group together related activity measurements
	UPPER_VALUE	NUMBER		Where the activity is a range, this represents the highest value of the range (numerically), while the PUBLISHED_VALUE column represents the lower value
	STANDARD_UPPER_VALUE	NUMBER		Where the activity is a range, this represents the standardised version of the highest value of the range (with the lower value represented by STANDARD_VALUE)
FK	SRC_ID	NUMBER		Foreign key to source table, indicating the source of the activity value
	TYPE	VARCHAR2(250)	NOT NULL	Type of end-point measurement: e.g. IC50, LD50, %inhibition etc, as it appears in the original dataset
	RELATION	VARCHAR2(50)		Symbol constraining the activity value (e.g. >, <, =), as it appears in the original dataset
	VALUE	NUMBER		Datapoint value as it appears in the original dataset.
	UNITS	VARCHAR2(100)		Units of measurement as they appear in the original dataset
	TEXT_VALUE	VARCHAR2(1000)		Non-numeric activities describing the overall assay outcome i.e. "Slightly active", "Not determined", "Toxic", "Active", "Inactive", "No effect"
	STANDARD_TEXT_VALUE	VARCHAR2(1000)		Standardized version of additional information about the measurement
FK	ACTION_TYPE	VARCHAR2(50)		Foreign key to action_type table; specifies the effect of the compound on its target.

ACTIVITY_PROPERTIES:

Table storing parameters and properties of Activity_IDs in the ACTIVITIES table - can be either independent variables that do not apply to the whole assay (e.g., compounds 'DOSE'), or dependent variables/results that are important in interpreting the activity values (e.g., 'HILL_SLOPE')

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	AP_ID	NUMBER	NOT NULL	Unique ID for each record.
FK,UK	ACTIVITY_ID	NUMBER	NOT NULL	FK to ACTIVITY_ID in ACTIVITIES table.
UK	TYPE	VARCHAR2(250)	NOT NULL	The parameter or property type
	RELATION	VARCHAR2(50)		Symbol constraining the value (e.g. >, <, =)
	VALUE	NUMBER		Numberical value for the parameter or property
	UNITS	VARCHAR2(100)		Units of measurement
	TEXT_VALUE	VARCHAR2(2000)		Non-numerical value of the parameter or property
	STANDARD_TYPE	VARCHAR2(250)		Standardised form of the TYPE
	STANDARD_RELATION	VARCHAR2(50)		Standardised form of the RELATION
	STANDARD_VALUE	NUMBER		Standardised form of the VALUE
	STANDARD_UNITS	VARCHAR2(100)		Standardised form of the UNITS
	STANDARD_TEXT_VALUE	VARCHAR2(2000)		Standardised form of the TEXT_VALUE
	COMMENTS	VARCHAR2(2000)		A Comment.
	RESULT_FLAG	NUMBER	NOT NULL	A flag to indicate, if set to 1, that this type is a dependent variable/result (e.g., slope) rather than an independent variable/parameter (0, the default).

ACTIVITY_SMID:

A join table between ACTIVITY_SUPP_MAP and ACTIVITY_SUPP

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	SMID	NUMBER	NOT NULL	FK to SMID in ACTIVITY_SUPP_MAP, and a FK to SMID in ACTIVITY_SUPP

ACTIVITY_STDS_LOOKUP:

Table storing details of standardised activity types and units, with permitted value ranges. Used to determine the correct standard_type and standard_units for a range of published types/units.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	STD_ACT_ID	NUMBER	NOT NULL	Primary key.
UK	STANDARD_TYPE	VARCHAR2(250)	NOT NULL	The standard_type that other published_types in the activities table have been converted to.
	DEFINITION	VARCHAR2(500)		A description/definition of the standard_type.
UK	STANDARD_UNITS	VARCHAR2(100)	NOT NULL	The units that are applied to this standard_type and to which other published_units are converted. Note a standard_type may have more than one allowable standard_unit and therefore multiple rows in this table.
	NORMAL_RANGE_MIN	NUMBER		The lowest value for this activity type that is likely to be genuine. This is only an approximation, so lower genuine values may exist, but it may be desirable to validate these before using them. For a given standard_type/units, values in the activities table below this threshold are flagged with a data_validity_comment of 'Outside typical range'.
	NORMAL_RANGE_MAX	NUMBER		The highest value for this activity type that is likely to be genuine. This is only an approximation, so higher genuine values may exist, but it may be desirable to validate these before using them. For a given standard_type/units, values in the activities table above this threshold are flagged with a data_validity_comment of 'Outside typical range'.

ACTIVITY_SUPP:

Supplementary / Supporting Data for ACTIVITIES - can be linked via ACTIVITY_SMID and ACTIVITY_SUPP_MAP tables

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	AS_ID	NUMBER	NOT NULL	Unique ID for each record.
UK	RGID	NUMBER	NOT NULL	Record Grouping ID, used to group together related data points in this table
FK	SMID	NUMBER		FK to SMID in ACTIVITY_SMID.
UK	TYPE	VARCHAR2(250)	NOT NULL	Type of end-point measurement: e.g. IC50, LD50, %inhibition etc, as it appears in the original dataset
	RELATION	VARCHAR2(50)		Symbol constraining the activity value (e.g. >, <, =), as it appears in the original dataset
	VALUE	NUMBER		Datapoint value as it appears in the original dataset.
	UNITS	VARCHAR2(100)		Units of measurement as they appear in the original dataset
	TEXT_VALUE	VARCHAR2(1000)		Non-numeric value for measurement as in original dataset
	STANDARD_TYPE	VARCHAR2(250)		Standardised form of the TYPE
	STANDARD_RELATION	VARCHAR2(50)		Standardised form of the RELATION
	STANDARD_VALUE	NUMBER		Standardised form of the VALUE
	STANDARD_UNITS	VARCHAR2(100)		Standardised form of the UNITS
	STANDARD_TEXT_VALUE	VARCHAR2(1000)		Standardised form of the TEXT_VALUE
	COMMENTS	VARCHAR2(4000)		A Comment.

ACTIVITY_SUPP_MAP:

Mapping table, linking supplementary / supporting data from the ACTIVITY_SUPP table to the main ACTIVITIES table

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ACTSM_ID	NUMBER	NOT NULL	Primary key
FK	ACTIVITY_ID	NUMBER	NOT NULL	FK to ACTIVITY_ID in ACTIVITIES table.
FK	SMID	NUMBER	NOT NULL	FK to SMID in ACTIVITY_SMID.

ASSAY_CLASS_MAP:

Mapping table linking assays to classes in the ASSAY_CLASSIFICATION table

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ASS_CLS_MAP_ID	NUMBER	NOT NULL	Primary key.
FK,UK	ASSAY_ID	NUMBER	NOT NULL	Foreign key that maps to the ASSAYS table
FK,UK	ASSAY_CLASS_ID	NUMBER	NOT NULL	Foreign key that maps to the ASSAY_CLASSIFICATION table

ASSAY_CLASSIFICATION:

Classification scheme for phenotypic assays e.g., by therapeutic area, phenotype/process and assay type. Can be used to find standard assays for a particular disease area or phenotype e.g., anti-obesity assays. Currently data are available only for in vivo efficacy assays

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ASSAY_CLASS_ID	NUMBER	NOT NULL	Primary key
	L1	VARCHAR2(100)		High level classification e.g., by anatomical/therapeutic area
	L2	VARCHAR2(100)		Mid-level classification e.g., by phenotype/biological process
UK	L3	VARCHAR2(1000)		Fine-grained classification e.g., by assay type
	CLASS_TYPE	VARCHAR2(50)		The type of assay being classified e.g., in vivo efficacy
	SOURCE	VARCHAR2(50)		Source from which the assay class was obtained

ASSAY_PARAMETERS:

Table storing additional parameters for an assay e.g., dose, administration route, time point

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ASSAY_PARAM_ID	NUMBER	NOT NULL	Numeric primary key
FK	ASSAY_ID	NUMBER	NOT NULL	Foreign key to assays table. The assay to which this parameter belongs
	TYPE	VARCHAR2(250)	NOT NULL	The type of parameter being described, according to the original data source
	RELATION	VARCHAR2(50)		The relation symbol for the parameter being described, according to the original data source
	VALUE	NUMBER		The value of the parameter being described, according to the original data source. Used for numeric data
	UNITS	VARCHAR2(100)		The units for the parameter being described, according to the original data source
	TEXT_VALUE	VARCHAR2(4000)		The text value of the parameter being described, according to the original data source. Used for non-numeric/qualitative data
	STANDARD_TYPE	VARCHAR2(250)		Standardized form of the TYPE
	STANDARD_RELATION	VARCHAR2(50)		Standardized form of the RELATION
	STANDARD_VALUE	NUMBER		Standardized form of the VALUE
	STANDARD_UNITS	VARCHAR2(100)		Standardized form of the UNITS
	STANDARD_TEXT_VALUE	VARCHAR2(4000)		Standardized form of the TEXT_VALUE
	COMMENTS	VARCHAR2(4000)		Additional comments describing the parameter

ASSAY_TYPE:

Description of assay types (e.g., Binding, Functional, ADMET)

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ASSAY_TYPE	VARCHAR2(1)	NOT NULL	Single character representing assay type
	ASSAY_DESC	VARCHAR2(250)		Description of assay type

ASSAYS:

Table storing a list of the assays that are reported in each document. Similar assays from different publications will appear as distinct assays in this table.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ASSAY_ID	NUMBER	NOT NULL	Unique ID for the assay
FK	DOC_ID	NUMBER	NOT NULL	Foreign key to documents table
	DESCRIPTION	VARCHAR2(4000)		Description of the reported assay
FK	ASSAY_TYPE	VARCHAR2(1)		Assay classification, e.g. B=Binding assay, A=ADME assay, F=Functional assay
	ASSAY_TEST_TYPE	VARCHAR2(20)		Type of assay system (i.e., in vivo or in vitro)
	ASSAY_CATEGORY	VARCHAR2(50)		screening, confirmatory (ie: dose-response), summary, panel, other,Thermal shift assay QC liability, Thermal shift assay, Affinity biochemical assay, Incucyte cell viability, Affinity phenotypic cellular assay, HTRF assay, Selectivity assay, Cell health data, NanoBRET assay, Alphascreen assay, Affinity on-target cellular assay, ITC assay, GPCR beta-arrestin recruitment assay
	ASSAY_ORGANISM	VARCHAR2(250)		Name of the organism for the assay system (e.g., the organism, tissue or cell line in which an assay was performed). May differ from the target organism (e.g., for a human protein expressed in non-human cells, or pathogen-infected human cells).
	ASSAY_TAX_ID	NUMBER		NCBI tax ID for the assay organism.
	ASSAY_STRAIN	VARCHAR2(200)		Name of specific strain of the assay organism used (where known)
	ASSAY_TISSUE	VARCHAR2(100)		Name of tissue used in the assay system (e.g., for tissue-based assays) or from which the assay system was derived (e.g., for cell/subcellular fraction-based assays).
	ASSAY_CELL_TYPE	VARCHAR2(100)		Name of cell type or cell line used in the assay system (e.g., for cell-based assays).
	ASSAY_SUBCELLULAR_FRACTION	VARCHAR2(100)		Name of subcellular fraction used in the assay system (e.g., microsomes, mitochondria).
FK	TID	NUMBER		Target identifier to which this assay has been mapped. Foreign key to target_dictionary. From ChEMBL_15 onwards, an assay will have only a single target assigned.
FK	RELATIONSHIP_TYPE	VARCHAR2(1)		Flag indicating of the relationship between the reported target in the source document and the assigned target from TARGET_DICTIONARY. Foreign key to RELATIONSHIP_TYPE table.
FK	CONFIDENCE_SCORE	NUMBER		Confidence score, indicating how accurately the assigned target(s) represents the actually assay target. Foreign key to CONFIDENCE_SCORE table. 0 means uncurated/unassigned, 1 = low confidence to 9 = high confidence.
FK	CURATED_BY	VARCHAR2(32)		Indicates the level of curation of the target assignment. Foreign key to curation_lookup table.
FK	SRC_ID	NUMBER	NOT NULL	Foreign key to source table
	SRC_ASSAY_ID	VARCHAR2(50)		Identifier for the assay in the source database/deposition (e.g., pubchem AID)
FK,UK	CHEMBL_ID	VARCHAR2(20)	NOT NULL	ChEMBL identifier for this assay (for use on web interface etc)
FK	CELL_ID	NUMBER		Foreign key to cell dictionary. The cell type or cell line used in the assay
FK	BAO_FORMAT	VARCHAR2(11)		ID for the corresponding format type in BioAssay Ontology (e.g., cell-based, biochemical, organism-based etc)
FK	TISSUE_ID	NUMBER		ID for the corresponding tissue/anatomy in Uberon. Foreign key to tissue_dictionary
FK	VARIANT_ID	NUMBER		Foreign key to variant_sequences table. Indicates the mutant/variant version of the target used in the assay (where known/applicable)
	AIDX	VARCHAR2(200)	NOT NULL	The Depositor Defined Assay Identifier
	ASSAY_GROUP	VARCHAR2(200)		Assays deposited across multiple datasets/releases that are considered comparable by the depositor are mapped to the same ASSAY GROUP. Allows depositors to group comparable assays and users to identify assays that can be considered identical.

ATC_CLASSIFICATION:

WHO ATC Classification for drugs

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
	WHO_NAME	VARCHAR2(2000)		WHO/INN name for the compound
	LEVEL1	VARCHAR2(10)		First level of classification
	LEVEL2	VARCHAR2(10)		Second level of classification
	LEVEL3	VARCHAR2(10)		Third level of classification
	LEVEL4	VARCHAR2(10)		Fourth level of classification
PK	LEVEL5	VARCHAR2(10)	NOT NULL	Complete ATC code for compound
	LEVEL1_DESCRIPTION	VARCHAR2(2000)		Description of first level of classification
	LEVEL2_DESCRIPTION	VARCHAR2(2000)		Description of second level of classification
	LEVEL3_DESCRIPTION	VARCHAR2(2000)		Description of third level of classification
	LEVEL4_DESCRIPTION	VARCHAR2(2000)		Description of fourth level of classification

BINDING_SITES:

Table storing details of binding sites for a target. A target may have multiple sites defined in this table.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	SITE_ID	NUMBER	NOT NULL	Primary key. Unique identifier for a binding site in a given target.
	SITE_NAME	VARCHAR2(200)		Name/label for the binding site.
FK	TID	NUMBER		Foreign key to target_dictionary. Target on which the binding site is found.

BIO_COMPONENT_SEQUENCES:

Table storing the sequences for biotherapeutic drugs (e.g., monoclonal antibodies, recombinant proteins, peptides etc. For multi-chain biotherapeutics (e.g., mAbs) each chain is stored here as a separate component.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	COMPONENT_ID	NUMBER	NOT NULL	Primary key. Unique identifier for each of the molecular components of biotherapeutics in ChEMBL (e.g., antibody chains, recombinant proteins, synthetic peptides).
	COMPONENT_TYPE	VARCHAR2(50)		Type of molecular component (e.g., 'Protein', 'Nucleic Acid'). The component type can be NULL for post-translational modifications such as the location of disulfide bridges, N-terminal glycosylation sites and C-termi lysine clipping.
	DESCRIPTION	VARCHAR2(200)		Description/name of molecular component.
	SEQUENCE	CLOB		Sequence of the biotherapeutic component.
	SEQUENCE_MD5SUM	VARCHAR2(32)		MD5 checksum of the sequence.
	TAX_ID	NUMBER		NCBI tax ID for the species from which the sequence is derived. May be null for humanized monoclonal antibodies, synthetic peptides etc.
	ORGANISM	VARCHAR2(150)		Name of the species from which the sequence is derived.

BIOASSAY_ONTOLOGY:

Lookup table providing labels for Bioassay Ontology terms used in assays and activities tables

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	BAO_ID	VARCHAR2(11)	NOT NULL	Bioassay Ontology identifier (BAO version 2.0)
	LABEL	VARCHAR2(100)	NOT NULL	Bioassay Ontology label for the term (BAO version 2.0)

BIOTHERAPEUTIC_COMPONENTS:

Links each biotherapeutic drug (in the biotherapeutics table) to its component sequences (in the bio_component_sequences table). A biotherapeutic drug can have multiple components and hence multiple rows in this table. Similarly, a particular component sequence can be par of more than one drug.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	BIOCOMP_ID	NUMBER	NOT NULL	Primary key.
FK,UK	MOLREGNO	NUMBER	NOT NULL	Foreign key to the biotherapeutics table, indicating which biotherapeutic the component is part of.
FK,UK	COMPONENT_ID	NUMBER	NOT NULL	Foreign key to the bio_component_sequences table, indicating which component is part of the biotherapeutic.

BIOTHERAPEUTICS:

Table mapping biotherapeutics (e.g. recombinant proteins, peptides and antibodies) to the molecule_dictionary table. Includes HELM notation where available.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK,FK	MOLREGNO	NUMBER	NOT NULL	Foreign key to molecule_dictionary
	DESCRIPTION	VARCHAR2(2000)		Description of the biotherapeutic.
	HELM_NOTATION	VARCHAR2(4000)		Sequence notation generated according to the HELM standard (http://www.openhelm.org/home). Currently for peptides only

CELL_DICTIONARY:

Table storing information for cell lines in the target_dictionary (e.g., tissue/species origin). Cell_name should match pref_name in target_dictionary.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	CELL_ID	NUMBER	NOT NULL	Primary key. Unique identifier for each cell line in the target_dictionary.
UK	CELL_NAME	VARCHAR2(50)	NOT NULL	Name of each cell line (as used in the target_dicitonary pref_name).
	CELL_DESCRIPTION	VARCHAR2(200)		Longer description (where available) of the cell line.
	CELL_SOURCE_TISSUE	VARCHAR2(50)		Tissue from which the cell line is derived, where known.
	CELL_SOURCE_ORGANISM	VARCHAR2(150)		Name of organism from which the cell line is derived.
UK	CELL_SOURCE_TAX_ID	NUMBER		NCBI tax ID of the organism from which the cell line is derived.
	CLO_ID	VARCHAR2(11)		ID for the corresponding cell line in Cell Line Ontology
	EFO_ID	VARCHAR2(12)		ID for the corresponding cell line in Experimental Factory Ontology
	CELLOSAURUS_ID	VARCHAR2(15)		ID for the corresponding cell line in Cellosaurus Ontology
	CL_LINCS_ID	VARCHAR2(8)		Cell ID used in LINCS (Library of Integrated Network-based Cellular Signatures)
FK,UK	CHEMBL_ID	VARCHAR2(20)		ChEMBL identifier for the cell (used in web interface etc)
	CELL_ONTOLOGY_ID	VARCHAR2(10)		ID for the corresponding cell type in the Cell Ontology

CHEMBL_ID_LOOKUP:

Lookup table storing chembl identifiers for different entities in the database (assays, compounds, documents and targets)

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	CHEMBL_ID	VARCHAR2(20)	NOT NULL	ChEMBL identifier
UK	ENTITY_TYPE	VARCHAR2(50)	NOT NULL	Type of entity (e.g., COMPOUND, ASSAY, TARGET)
UK	ENTITY_ID	NUMBER	NOT NULL	Primary key for that entity in corresponding table (e.g., molregno for compounds, tid for targets)
	STATUS	VARCHAR2(10)	NOT NULL	Indicates whether the status of the entity within the database - ACTIVE, INACTIVE (downgraded), OBS (obsolete/removed).
	LAST_ACTIVE	NUMBER		indicates the last ChEMBL version where the CHEMBL_ID was active

CHEMBL_RELEASE:

Table listing ChEMBL releases with their creation dates

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	CHEMBL_RELEASE_ID	INTEGER	NOT NULL	Primary key
	CHEMBL_RELEASE	VARCHAR2(20)		ChEMBL release name
	CREATION_DATE	DATE		ChEMBL release creation date

COMPONENT_CLASS:

Links protein components of targets to the protein_family_classification table. A protein can have more than one classification (e.g., Membrane receptor and Enzyme).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
FK,UK	COMPONENT_ID	NUMBER	NOT NULL	Foreign key to component_sequences table.
FK,UK	PROTEIN_CLASS_ID	NUMBER	NOT NULL	Foreign key to the protein_classification table.
PK	COMP_CLASS_ID	NUMBER	NOT NULL	Primary key.

COMPONENT_DOMAINS:

Links protein components of targets to the structural domains they contain (from the domains table). Contains information showing the start and end position of the domain in the component sequence.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	COMPD_ID	NUMBER	NOT NULL	Primary key.
FK,UK	DOMAIN_ID	NUMBER		Foreign key to the domains table, indicating the domain that is contained in the associated molecular component.
FK,UK	COMPONENT_ID	NUMBER	NOT NULL	Foreign key to the component_sequences table, indicating the molecular_component that has the given domain.
UK	START_POSITION	NUMBER		Start position of the domain within the sequence given in the component_sequences table.
	END_POSITION	NUMBER		End position of the domain within the sequence given in the component_sequences table.

COMPONENT_GO:

Table mapping protein components in the COMPONENT_SEQUENCES table to the GO slim terms stored in the GO_CLASSIFICATION table

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	COMP_GO_ID	NUMBER	NOT NULL	Primary key
FK,UK	COMPONENT_ID	NUMBER	NOT NULL	Foreign key to COMPONENT_SEQUENCES table. The protein component this GO term applies to
FK,UK	GO_ID	VARCHAR2(10)	NOT NULL	Foreign key to the GO_CLASSIFICATION table. The GO term that this protein is mapped to

COMPONENT_SEQUENCES:

Table storing the sequences for components of molecular targets (e.g., protein sequences), along with other details taken from sequence databases (e.g., names, accessions). Single protein targets will have a single protein component in this table, whereas protein complexes/protein families will have multiple protein components.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	COMPONENT_ID	NUMBER	NOT NULL	Primary key. Unique identifier for the component.
	COMPONENT_TYPE	VARCHAR2(50)		Type of molecular component represented (e.g., 'PROTEIN','DNA','RNA').
UK	ACCESSION	VARCHAR2(25)		Accession for the sequence in the source database from which it was taken (e.g., UniProt accession for proteins).
	SEQUENCE	CLOB		A representative sequence for the molecular component, as given in the source sequence database (not necessarily the exact sequence used in the assay).
	SEQUENCE_MD5SUM	VARCHAR2(32)		MD5 checksum of the sequence.
	DESCRIPTION	VARCHAR2(200)		Description/name for the molecular component, usually taken from the source sequence database.
	TAX_ID	NUMBER		NCBI tax ID for the sequence in the source database (i.e., species that the protein/nucleic acid sequence comes from).
	ORGANISM	VARCHAR2(150)		Name of the organism the sequence comes from.
	DB_SOURCE	VARCHAR2(25)		The name of the source sequence database from which sequences/accessions are taken. For UniProt proteins, this field indicates whether the sequence is from SWISS-PROT or TREMBL.
	DB_VERSION	VARCHAR2(10)		The version of the source sequence database from which sequences/accession were last updated.

COMPONENT_SYNONYMS:

Table storing synonyms for the components of molecular targets (e.g., names, acronyms, gene symbols etc.) Please note: EC numbers are also currently included in this table although they are not strictly synonyms and can apply to multiple proteins.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	COMPSYN_ID	NUMBER	NOT NULL	Primary key.
FK,UK	COMPONENT_ID	NUMBER	NOT NULL	Foreign key to the component_sequences table. The component to which this synonym applies.
UK	COMPONENT_SYNONYM	VARCHAR2(500)		The synonym for the component.
UK	SYN_TYPE	VARCHAR2(20)		The type or origin of the synonym (e.g., GENE_SYMBOL).

COMPOUND_PROPERTIES:

Table storing calculated physicochemical properties for compounds, now calculated with RDKit and ChemAxon software (note all but FULL_MWT and FULL_MOLFORMULA are calculated on the parent structure)

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK,FK	MOLREGNO	NUMBER	NOT NULL	Foreign key to compounds table (compound structure)
	MW_FREEBASE	NUMBER		Molecular weight of parent compound
	ALOGP	NUMBER		Calculated ALogP
	HBA	NUMBER		Number hydrogen bond acceptors
	HBD	NUMBER		Number hydrogen bond donors
	PSA	NUMBER		Polar surface area
	RTB	NUMBER		Number rotatable bonds
	RO3_PASS	VARCHAR2(3)		Indicates whether the compound passes the rule-of-three (mw < 300, logP < 3 etc)
	NUM_RO5_VIOLATIONS	NUMBER		Number of violations of Lipinski's rule-of-five, using HBA and HBD definitions
	FULL_MWT	NUMBER		Molecular weight of the full compound including any salts
	AROMATIC_RINGS	NUMBER		Number of aromatic rings
	HEAVY_ATOMS	NUMBER		Number of heavy (non-hydrogen) atoms
	QED_WEIGHTED	NUMBER		Weighted quantitative estimate of drug likeness (as defined by Bickerton et al., Nature Chem 2012)
	FULL_MOLFORMULA	VARCHAR2(100)		Molecular formula for the full compound (including any salt)
	NP_LIKENESS_SCORE	NUMBER		Natural Product-likeness Score: Peter Ertl, Silvio Roggo, and Ansgar Schuffenhauer Journal of Chemical Information and Modeling, 48, 68-74 (2008)

COMPOUND_RECORDS:

Represents each compound extracted from scientific documents.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	RECORD_ID	NUMBER	NOT NULL	Unique ID for a compound/record
FK	MOLREGNO	NUMBER		Foreign key to compounds table (compound structure)
FK	DOC_ID	NUMBER	NOT NULL	Foreign key to documents table
	COMPOUND_KEY	VARCHAR2(250)		Key text identifying this compound in the scientific document
	COMPOUND_NAME	VARCHAR2(4000)		Name of this compound recorded in the scientific document
FK	SRC_ID	NUMBER	NOT NULL	Foreign key to source table
	SRC_COMPOUND_ID	VARCHAR2(150)		Identifier for the compound in the source database (e.g., pubchem SID)
	CIDX	VARCHAR2(200)	NOT NULL	The Depositor Defined Compound Identifier.

COMPOUND_STRUCTURAL_ALERTS:

Table showing which structural alerts (as defined in the STRUCTURAL_ALERTS table) are found in a particular ChEMBL compound. It should be noted some alerts/alert sets are more permissive than others and may flag a large number of compounds. Results should be interpreted with care, depending on the use-case, and not treated as a blanket filter (e.g., around 50% of approved drugs have 1 or more alerts from these sets).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	CPD_STR_ALERT_ID	NUMBER	NOT NULL	Primary key.
FK,UK	MOLREGNO	NUMBER	NOT NULL	Foreign key to the molecule_dictionary. The compound for which the structural alert has been found.
FK,UK	ALERT_ID	NUMBER	NOT NULL	Foreign key to the structural_alerts table. The particular alert that has been identified in this compound.

COMPOUND_STRUCTURES:

Table storing various structure representations (e.g., Molfile, InChI) for each compound

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK,FK	MOLREGNO	NUMBER	NOT NULL	Internal Primary Key for the compound structure and foreign key to molecule_dictionary table
	MOLFILE	CLOB		MDL Connection table representation of compound
UK	STANDARD_INCHI	VARCHAR2(4000)		IUPAC standard InChI for the compound
UK	STANDARD_INCHI_KEY	VARCHAR2(27)	NOT NULL	IUPAC standard InChI key for the compound
	CANONICAL_SMILES	VARCHAR2(4000)		Canonical smiles, generated using RDKit

CONFIDENCE_SCORE_LOOKUP:

Lookup table describing how assays confidence scores are assigned depending on the type of target(s) assigned to the assay and the level of confidence in their molecular identity

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	CONFIDENCE_SCORE	NUMBER	NOT NULL	0-9 score showing level of confidence in assignment of the precise molecular target of the assay
	DESCRIPTION	VARCHAR2(100)	NOT NULL	Description of the target types assigned with each score
	TARGET_MAPPING	VARCHAR2(30)	NOT NULL	Short description of the target types assigned with each score

CURATION_LOOKUP:

Lookup table for assays.curated_by column. Shows level of curation that has been applied to the assay to target mapping.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	CURATED_BY	VARCHAR2(32)	NOT NULL	Short description of the level of curation
	DESCRIPTION	VARCHAR2(100)	NOT NULL	Definition of terms in the curated_by field.

DATA_VALIDITY_LOOKUP:

Table storing information about the data_validity_comment values used in the activities table.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	DATA_VALIDITY_COMMENT	VARCHAR2(30)	NOT NULL	Primary key. Short description of various types of errors/warnings applied to values in the activities table.
	DESCRIPTION	VARCHAR2(200)		Definition of the terms in the data_validity_comment field.

DEFINED_DAILY_DOSE:

WHO DDD (defined daily dose) information

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
FK	ATC_CODE	VARCHAR2(10)	NOT NULL	ATC code for the compound (foreign key to ATC_CLASSIFICATION table)
	DDD_UNITS	VARCHAR2(200)		Units of defined daily dose
	DDD_ADMR	VARCHAR2(1000)		Administration route for dose
	DDD_COMMENT	VARCHAR2(2000)		Comment
PK	DDD_ID	NUMBER	NOT NULL	Internal primary key
	DDD_VALUE	NUMBER		Value of defined daily dose

DOCS:

Holds all scientific documents (journal articles or patents) from which assays have been extracted.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	DOC_ID	NUMBER	NOT NULL	Unique ID for the document
	JOURNAL	VARCHAR2(50)		Abbreviated journal name for an article
	YEAR	NUMBER		Year of journal article publication
	VOLUME	VARCHAR2(50)		Volume of journal article
	ISSUE	VARCHAR2(50)		Issue of journal article
	FIRST_PAGE	VARCHAR2(50)		First page number of journal article
	LAST_PAGE	VARCHAR2(50)		Last page number of journal article
	PUBMED_ID	NUMBER		NIH pubmed record ID, where available
	DOI	VARCHAR2(100)		Digital object identifier for this reference
FK,UK	CHEMBL_ID	VARCHAR2(20)	NOT NULL	ChEMBL identifier for this document (for use on web interface etc)
	TITLE	VARCHAR2(500)		Document title (e.g., Publication title or description of dataset)
	DOC_TYPE	VARCHAR2(50)	NOT NULL	Type of the document (e.g., Publication, Deposited dataset)
	AUTHORS	VARCHAR2(4000)		For a deposited dataset, the authors carrying out the screening and/or submitting the dataset.
	ABSTRACT	CLOB		For a deposited dataset, a brief description of the dataset.
	PATENT_ID	VARCHAR2(20)		Patent ID for this document
	RIDX	VARCHAR2(200)	NOT NULL	The Depositor Defined Reference Identifier
FK	SRC_ID	INTEGER	NOT NULL	Foreign key to Source table, indicating the source of this document
FK	CHEMBL_RELEASE_ID	INTEGER		Foreign key to chembl_release table
	CONTACT	VARCHAR2(200)		A stable identifier (ideally ORCID) for one to three (stable) primary contact person(s) for this data set should be provided; this could be the individual carrying out the experiment, the head of a laboratory or a line manager.

DOMAINS:

Table storing a non-redundant list of domains found in protein targets (e.g., Pfam domains).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	DOMAIN_ID	NUMBER	NOT NULL	Primary key. Unique identifier for each domain.
	DOMAIN_TYPE	VARCHAR2(20)	NOT NULL	Indicates the source of the domain (e.g., Pfam).
	SOURCE_DOMAIN_ID	VARCHAR2(20)	NOT NULL	Identifier for the domain in the source database (e.g., Pfam ID such as PF00001).
	DOMAIN_NAME	VARCHAR2(100)		Name given to the domain in the source database (e.g., 7tm_1).
	DOMAIN_DESCRIPTION	VARCHAR2(500)		Longer name or description for the domain.

DRUG_INDICATION:

Table storing indications for drugs, and clinical candidate drugs, from a variety of sources (e.g., FDA, EMA, WHO ATC, ClinicalTrials.gov, INN, USAN).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	DRUGIND_ID	NUMBER	NOT NULL	Primary key
FK,UK	RECORD_ID	NUMBER	NOT NULL	Foreign key to compound_records table. Links to the drug record to which this indication applies
FK	MOLREGNO	NUMBER		Molregno for the drug (foreign key to the molecule_dictionary and compound_records tables)
	MAX_PHASE_FOR_IND	NUMBER		Maximum phase of development that the drug is known to have reached for this particular indication (4 = Approved, 3 = Phase 3 Clinical Trials, 2 = Phase 2 Clinical Trials, 1 = Phase 1 Clinical Trials, 0.5 = Early Phase 1 Clinical Trials, -1 = Clinical Phase unknown for drug or clinical candidate drug ie where ChEMBL cannot assign a clinical phase)
UK	MESH_ID	VARCHAR2(20)	NOT NULL	Medical Subject Headings (MeSH) disease identifier corresponding to the indication
	MESH_HEADING	VARCHAR2(200)	NOT NULL	Medical Subject Heading term for the MeSH disease ID
UK	EFO_ID	VARCHAR2(20)		Experimental Factor Ontology (EFO) disease identifier corresponding to the indication
	EFO_TERM	VARCHAR2(200)		Experimental Factor Ontology term for the EFO ID

DRUG_MECHANISM:

Table storing mechanism of action information for drugs, and clinical candidate drugs, from a variety of sources (e.g., ATC, FDA, ClinicalTrials.gov)

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	MEC_ID	NUMBER	NOT NULL	Primary key for each drug mechanism of action
FK	RECORD_ID	NUMBER	NOT NULL	Record_id for the drug (foreign key to compound_records table)
FK	MOLREGNO	NUMBER		Molregno for the drug (foreign key to molecule_dictionary table)
	MECHANISM_OF_ACTION	VARCHAR2(250)		Description of the mechanism of action e.g., 'Phosphodiesterase 5 inhibitor'
FK	TID	NUMBER		Target associated with this mechanism of action (foreign key to target_dictionary table)
FK	SITE_ID	NUMBER		Binding site for the drug within the target (where known) - foreign key to binding_sites table
FK	ACTION_TYPE	VARCHAR2(50)		Type of action of the drug on the target e.g., agonist/antagonist etc (foreign key to action_type table)
	DIRECT_INTERACTION	NUMBER		Flag to show whether the molecule is believed to interact directly with the target (1 = yes, 0 = no)
	MOLECULAR_MECHANISM	NUMBER		Flag to show whether the mechanism of action describes the molecular target of the drug, rather than a higher-level physiological mechanism e.g., vasodilator (1 = yes, 0 = no)
	DISEASE_EFFICACY	NUMBER		Flag to show whether the target assigned is believed to play a role in the efficacy of the drug in the indication(s) for which it is approved (1 = yes, 0 = no)
	MECHANISM_COMMENT	VARCHAR2(2000)		Additional comments regarding the mechanism of action
	SELECTIVITY_COMMENT	VARCHAR2(1000)		Additional comments regarding the selectivity of the drug
	BINDING_SITE_COMMENT	VARCHAR2(1000)		Additional comments regarding the binding site of the drug
FK	VARIANT_ID	NUMBER		Foreign key to variant_sequences table. Indicates the mutant/variant version of the target used in the assay (where known/applicable)

DRUG_WARNING:

Table storing safety-related information for drugs and clinical candidates

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	WARNING_ID	NUMBER	NOT NULL	Primary key for the drug warning
FK	RECORD_ID	NUMBER		Foreign key to the compound_records table
	MOLREGNO	NUMBER		Foreign key to molecule_dictionary table
	WARNING_TYPE	VARCHAR2(20)		Description of the drug warning type (e.g., withdrawn vs black box warning)
	WARNING_CLASS	VARCHAR2(100)		High-level class of the drug warning
	WARNING_DESCRIPTION	VARCHAR2(4000)		Description of the drug warning
	WARNING_COUNTRY	VARCHAR2(1000)		List of countries/regions associated with the drug warning
	WARNING_YEAR	NUMBER		Earliest year the warning was applied to the drug.
	EFO_TERM	VARCHAR2(200)		Term for Experimental Factor Ontology (EFO)
	EFO_ID	VARCHAR2(20)		Identifier for Experimental Factor Ontology (EFO)
	EFO_ID_FOR_WARNING_CLASS	VARCHAR2(20)		Warning Class Identifier for Experimental Factor Ontology (EFO)

FORMULATIONS:

Table linking individual ingredients in approved products (and their strengths) to entries in the molecule dictionary. Where products are mixtures of active ingredients they will have multiple entries in this table

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
FK,UK	PRODUCT_ID	VARCHAR2(30)	NOT NULL	Unique identifier of the product. FK to PRODUCTS
	INGREDIENT	VARCHAR2(200)		Name of the approved ingredient within the product
	STRENGTH	VARCHAR2(300)		Dose strength
FK,UK	RECORD_ID	NUMBER	NOT NULL	Foreign key to the compound_records table.
FK	MOLREGNO	NUMBER		Unique identifier of the ingredient FK to MOLECULE_DICTIONARY
PK	FORMULATION_ID	NUMBER	NOT NULL	Primary key.

GO_CLASSIFICATION:

Table storing the ChEMBL Drug Target GO slim (http://www.geneontology.org/ontology/subsets/goslim_chembl.obo)

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	GO_ID	VARCHAR2(10)	NOT NULL	Primary key. Gene Ontology identifier for the GO slim term
	PARENT_GO_ID	VARCHAR2(10)		Gene Ontology identifier for the parent of this GO term in the ChEMBL Drug Target GO slim
	PREF_NAME	VARCHAR2(200)		Gene Ontology name
	CLASS_LEVEL	NUMBER		Indicates the level of the term in the slim (L1 = highest)
	ASPECT	VARCHAR2(1)		Indicates which aspect of the Gene Ontology the term belongs to (F = molecular function, P = biological process, C = cellular component)
	PATH	VARCHAR2(1000)		Indicates the full path to this term in the GO slim

INDICATION_REFS:

Table storing references indicating the source of drug indication information

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	INDREF_ID	NUMBER	NOT NULL	Primary key
FK,UK	DRUGIND_ID	NUMBER	NOT NULL	Foreign key to the DRUG_INDICATION table, indicating the drug-indication link that this reference applies to
UK	REF_TYPE	VARCHAR2(50)	NOT NULL	Type/source of reference
UK	REF_ID	VARCHAR2(4000)	NOT NULL	Identifier for the reference in the source
	REF_URL	VARCHAR2(4000)	NOT NULL	Full URL linking to the reference

LIGAND_EFF:

Contains BEI (Binding Efficiency Index) and SEI (Surface Binding Efficiency Index) for each activity_id where such data can be calculated.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK,FK	ACTIVITY_ID	NUMBER	NOT NULL	Link key to activities table
	BEI	NUMBER		Binding Efficiency Index = p(XC50) *1000/MW_freebase
	SEI	NUMBER		Surface Efficiency Index = p(XC50)*100/PSA
	LE	NUMBER		Ligand Efficiency = deltaG/heavy_atoms [from the Hopkins DDT paper 2004]
	LLE	NUMBER		Lipophilic Ligand Efficiency = -logKi-ALogP. [from Leeson NRDD 2007]

MECHANISM_REFS:

Table storing references for information in the drug_mechanism table

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	MECREF_ID	NUMBER	NOT NULL	Primary key
FK,UK	MEC_ID	NUMBER	NOT NULL	Foreign key to drug_mechanism table - indicating the mechanism to which the references refer
UK	REF_TYPE	VARCHAR2(50)	NOT NULL	Type/source of reference (e.g., 'PubMed','DailyMed')
UK	REF_ID	VARCHAR2(400)		Identifier for the reference in the source (e.g., PubMed ID or DailyMed setid)
	REF_URL	VARCHAR2(400)		Full URL linking to the reference

METABOLISM:

Table storing drug metabolic pathways, manually curated from a variety of sources

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	MET_ID	NUMBER	NOT NULL	Primary key
FK,UK	DRUG_RECORD_ID	NUMBER		Foreign key to compound_records. Record representing the drug or other compound for which metabolism is being studied (may not be the same as the substrate being measured)
FK,UK	SUBSTRATE_RECORD_ID	NUMBER		Foreign key to compound_records. Record representing the compound that is the subject of metabolism
FK,UK	METABOLITE_RECORD_ID	NUMBER		Foreign key to compound_records. Record representing the compound that is the result of metabolism
UK	PATHWAY_ID	NUMBER		Identifier for the metabolic scheme/pathway (may be multiple pathways from one source document)
	PATHWAY_KEY	VARCHAR2(50)		Link to original source indicating where the pathway information was found (e.g., Figure 1, page 23)
UK	ENZYME_NAME	VARCHAR2(200)		Name of the enzyme responsible for the metabolic conversion
FK,UK	ENZYME_TID	NUMBER		Foreign key to target_dictionary. TID for the enzyme responsible for the metabolic conversion
	MET_CONVERSION	VARCHAR2(200)		Description of the metabolic conversion
	ORGANISM	VARCHAR2(100)		Organism in which this metabolic reaction occurs
UK	TAX_ID	NUMBER		NCBI Tax ID for the organism in which this metabolic reaction occurs
	MET_COMMENT	VARCHAR2(1000)		Additional information regarding the metabolism (e.g., organ system, conditions under which observed, activity of metabolites)

METABOLISM_REFS:

Table storing references for metabolic pathways, indicating the source of the data

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	METREF_ID	NUMBER	NOT NULL	Primary key
FK,UK	MET_ID	NUMBER	NOT NULL	Foreign key to record_metabolism table - indicating the metabolism information to which the references refer
UK	REF_TYPE	VARCHAR2(50)	NOT NULL	Type/source of reference (e.g., 'PubMed','DailyMed')
UK	REF_ID	VARCHAR2(200)		Identifier for the reference in the source (e.g., PubMed ID or DailyMed setid)
	REF_URL	VARCHAR2(400)		Full URL linking to the reference

MOLECULE_ATC_CLASSIFICATION:

Table mapping drugs in the molecule_dictionary to ATC codes in the atc_classification table

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	MOL_ATC_ID	NUMBER	NOT NULL	Primary key
FK	LEVEL5	VARCHAR2(10)	NOT NULL	ATC code (foreign key to atc_classification table)
FK	MOLREGNO	NUMBER	NOT NULL	Drug to which the ATC code applies (foreign key to molecule_dictionary table)

MOLECULE_DICTIONARY:

Table storing a non-redundant list of curated compounds for ChEMBL (includes preclinical compounds, drugs and clinical candidate drugs) and some associated attributes.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	MOLREGNO	NUMBER	NOT NULL	Internal Primary Key for the molecule
	PREF_NAME	VARCHAR2(255)		Preferred name for the molecule
FK,UK	CHEMBL_ID	VARCHAR2(20)	NOT NULL	ChEMBL identifier for this compound (for use on web interface etc)
	MAX_PHASE	NUMBER		Maximum phase of development reached for the compound across all indications (4 = Approved, 3 = Phase 3 Clinical Trials, 2 = Phase 2 Clinical Trials, 1 = Phase 1 Clinical Trials, 0.5 = Early Phase 1 Clinical Trials, -1 = Clinical Phase unknown for drug or clinical candidate drug ie where ChEMBL cannot assign a clinical phase, NULL = preclinical compounds with bioactivity data)
	THERAPEUTIC_FLAG	NUMBER	NOT NULL	Indicates that a drug has a therapeutic application, as opposed to e.g., an imaging agent, additive etc (1 = yes, 0 = default value).
	DOSED_INGREDIENT	NUMBER	NOT NULL	Indicates that the drug is dosed in this form, e.g., a particular salt (1 = yes, 0 = default value)
	STRUCTURE_TYPE	VARCHAR2(10)	NOT NULL	Indicates whether the molecule has a small molecule structure or a protein sequence (MOL indicates an entry in the compound_structures table, SEQ indications an entry in the protein_therapeutics table, NON indicates an entry in neither table, e.g., structure unknown)
	MOLECULE_TYPE	VARCHAR2(30)		Type of molecule (Small molecule, Protein, Antibody, Antibody drug conjugate, Oligosaccharide, Oligonucleotide, Cell, Enzyme, Gene, Unknown)
	FIRST_APPROVAL	NUMBER		Earliest known approval year for the drug (NULL is the default value)
	ORAL	NUMBER	NOT NULL	Indicates whether the drug is known to be administered orally (1 = yes, 0 = default value)
	PARENTERAL	NUMBER	NOT NULL	Indicates whether the drug is known to be administered parenterally (1 = yes, 0 = default value)
	TOPICAL	NUMBER	NOT NULL	Indicates whether the drug is known to be administered topically (1 = yes, 0 = default value).
	BLACK_BOX_WARNING	NUMBER	NOT NULL	Indicates that the drug has a black box warning (1 = yes, 0 = default value)
	NATURAL_PRODUCT	NUMBER	NOT NULL	Indicates whether the compound is a natural product as defined by COCONUT (https://coconut.naturalproducts.net/), the COlleCtion of Open Natural ProdUcTs. (1 = yes, 0 = default value)
	FIRST_IN_CLASS	NUMBER	NOT NULL	Indicates whether this is known to be the first approved drug of its class (e.g., acting on a particular target). This is regardless of the indication, or the route of administration (1 = yes, 0 = no, -1 = preclinical compound ie not a drug).
	CHIRALITY	NUMBER	NOT NULL	Indicates the chirality of the drug (2 = an achiral molecule, 1 = single enantiomer where all stereocenters have known absolute configuration, 0 = a mixture of stereoisomers such as a racemic mixture, epimeric mixture, etc, -1 = has unknown chirality)
	PRODRUG	NUMBER	NOT NULL	Indicates that the drug is a pro-drug. See active_molregno field in molecule hierarchy for the pharmacologically active molecule, where known (1 = yes, 0 = no, -1 = preclinical compound ie not a drug)
	INORGANIC_FLAG	NUMBER	NOT NULL	Indicates whether the molecule is inorganic i.e., containing only metal atoms and <2 carbon atoms (1 = yes, 0 = no, -1 = preclinical compound ie not a drug)
	USAN_YEAR	NUMBER		The year in which the application for a USAN name was granted. (NULL is the default value)
	AVAILABILITY_TYPE	NUMBER		The availability type for the drug (-2 = withdrawn, -1 = unknown, 0 = discontinued, 1 = prescription only, 2 = over the counter)
	USAN_STEM	VARCHAR2(50)		Where the drug or clinical candidate name can be matched, this indicates the USAN stem (NULL is the default value). Also described in the USAN_STEMS table.
	POLYMER_FLAG	NUMBER		Indicates whether a molecule is a small molecule polymer, e.g., polystyrex (1 = yes, 0 = default value)
	USAN_SUBSTEM	VARCHAR2(50)		Where the drug or clinical candidate name can be matched, this indicates the USAN substem (NULL is the default value)
	USAN_STEM_DEFINITION	VARCHAR2(1000)		Definition of the USAN stem (NULL is the default value)
	WITHDRAWN_FLAG	NUMBER	NOT NULL	Indicates an approved drug has been withdrawn for toxicity reasons for all indications, for all populations at all doses in at least one country (not necessarily in the US). (1 = yes, 0 = default value)
	CHEMICAL_PROBE	NUMBER	NOT NULL	Indicates whether the compound is a chemical probe; for exact definition see release notes (1 = yes, 0 = default value).
	ORPHAN	NUMBER	NOT NULL	Indicates orphan designation, i.e. intended for use against a rare condition (1 = yes, 0 = no, -1 = preclinical compound ie not a drug)
	VETERINARY	NUMBER		Indicates whether the drug is known to have a veterinary product. A veterinary product may also be a medicinal product for Human use. (1 = yes, 0 = no, -1 = preclinical compound ie not a drug).

MOLECULE_HIERARCHY:

Table storing relationships between parents, salts and active metabolites (for pro-drugs).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK,FK	MOLREGNO	NUMBER	NOT NULL	Foreign key to compounds table. This field holds a list of all of the ChEMBL compounds with associated data (e.g., activity information, approved drugs). Parent compounds that are generated only by removing salts, and which do not themselves have any associated data will not appear here.
FK	PARENT_MOLREGNO	NUMBER		Represents parent compound of molregno in first field (i.e., generated by removing salts). Where molregno and parent_molregno are same, the initial ChEMBL compound did not contain a salt component, or else could not be further processed for various reasons (e.g., inorganic mixture). Compounds which are only generated by removing salts will appear in this field only. Those which, themselves, have any associated data (e.g., activity data) or are launched drugs will also appear in the molregno field.
FK	ACTIVE_MOLREGNO	NUMBER		Where a compound is a pro-drug, this represents the active metabolite of the 'dosed' compound given by parent_molregno. Where parent_molregno and active_molregno are the same, the compound is not currently known to be a pro-drug.

MOLECULE_SYNONYMS:

Stores synonyms for a compound (e.g., common names, trade names, research codes etc)

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
FK,UK	MOLREGNO	NUMBER	NOT NULL	Foreign key to molecule_dictionary
UK	SYN_TYPE	VARCHAR2(50)	NOT NULL	Type of name/synonym (e.g., TRADE_NAME, RESEARCH_CODE, USAN)
PK	MOLSYN_ID	NUMBER	NOT NULL	Primary key.
UK	SYNONYMS	VARCHAR2(250)		Synonym for the compound

ORGANISM_CLASS:

Simple organism classification (essentially a cut-down version of the NCBI taxonomy for organisms in ChEMBL target_dictionary table), allowing browsing of ChEMBL data by taxonomic groups

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	OC_ID	NUMBER	NOT NULL	Internal primary key
UK	TAX_ID	NUMBER		NCBI taxonomy ID for the organism (corresponding to tax_ids in target_dictionary table)
	L1	VARCHAR2(200)		Highest level classification (e.g., Eukaryotes, Bacteria, Fungi etc)
	L2	VARCHAR2(200)		Second level classification
	L3	VARCHAR2(200)		Third level classification

PATENT_USE_CODES:

Table from FDA Orange Book, showing definitions of different patent use codes (as used in the product_patents table).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	PATENT_USE_CODE	VARCHAR2(8)	NOT NULL	Primary key. Patent use code from FDA Orange Book
	DEFINITION	VARCHAR2(500)	NOT NULL	Definition for the patent use code, from FDA Orange Book.

PESTICIDE_CLASS_MAPPING:

Table mapping compounds in the PESTICIDE CLASSIFICATION table to the MOLECULE_DICTIONARY table.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	MOL_PEST_ID	NUMBER	NOT NULL	Primary key for the PESTICIDE_CLASS_MAPPING table
FK,UK	PEST_CLASS_ID	NUMBER		Foreign key to the PESTICIDE_CLASSIFICATION table
FK,UK	MOLREGNO	NUMBER		ChEMBL molecule registration number. Foreign key to the MOLECULE_DICTIONARY table

PESTICIDE_CLASSIFICATION:

Table describing the classification of the pesticide mechanism of action according to the Fungicide Resistance Action Committee (FRAC), Herbicide Resistance Action Committee (HRAC), and Insecticide Resistance Action Committee (IRAC).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	PEST_CLASS_ID	NUMBER	NOT NULL	Primary key for the PESTICIDE_CLASSIFICATION table
UK	COMPOUND_NAME	VARCHAR2(2000)		Name used for the compound in the classification by FRAC, HRAC and IRAC. Use the COMPOUND_NAME field in conjunction with the REF_ID and REF_URL fields to identify the appropriate row within the source file.
FK,UK	MEC_ID	NUMBER		The identifier for the mechanism-of-action. Foreign key to the DRUG_MECHANISM table.
UK	MECHANISM_COMMENT	VARCHAR2(2000)		Additional mechanism information about the classification, as described by FRAC, HRAC and IRAC.
UK	REF_TYPE	VARCHAR2(50)		The source of the pesticide classification (Fungicide Resistance Action Committee (FRAC), Herbicide Resistance Action Committee (HRAC), or Insecticide Resistance Action Committee (IRAC).
	REF_ID	VARCHAR2(400)		The name of the file from which the classification is derived. The location of the file is given in the REF_URL field. Use the name given in the COMPOUND_NAME field to identify the row within the file.
	REF_URL	VARCHAR2(400)		Full URL for the source information.

PREDICTED_BINDING_DOMAINS:

Table storing information on the likely binding domain of compounds in the activities table (based on analysis of the domain structure of the target. Note these are predictions, not experimentally determined. See Kruger F, Rostom R and Overington JP (2012), BMC Bioinformatics 13(S17), S11 for more details.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	PREDBIND_ID	NUMBER	NOT NULL	Primary key.
FK	ACTIVITY_ID	NUMBER		Foreign key to the activities table, indicating the compound/assay(+target) combination for which this prediction is made.
FK	SITE_ID	NUMBER		Foreign key to the binding_sites table, indicating the binding site (domain) that the compound is predicted to bind to.
	PREDICTION_METHOD	VARCHAR2(50)		The method used to assign the binding domain (e.g., 'Single domain' where the protein has only 1 domain, 'Multi domain' where the protein has multiple domains, but only 1 is known to bind small molecules in other proteins).
	CONFIDENCE	VARCHAR2(10)		The level of confidence assigned to the prediction (high where the protein has only 1 domain, medium where the compound has multiple domains, but only 1 known small molecule-binding domain).

PRODUCT_PATENTS:

Table from FDA Orange Book, showing patents associated with drug products.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	PROD_PAT_ID	NUMBER	NOT NULL	Primary key
FK,UK	PRODUCT_ID	VARCHAR2(30)	NOT NULL	Foreign key to products table - FDA application number for the product
UK	PATENT_NO	VARCHAR2(20)	NOT NULL	Patent numbers as submitted by the applicant holder for patents covered by the statutory provisions
UK	PATENT_EXPIRE_DATE	DATE	NOT NULL	Date the patent expires as submitted by the applicant holder including applicable extensions
	DRUG_SUBSTANCE_FLAG	NUMBER	NOT NULL	Patents submitted on FDA Form 3542 and listed after August 18, 2003 may have a drug substance flag set to 1, indicating the sponsor submitted the patent as claiming the drug substance
	DRUG_PRODUCT_FLAG	NUMBER	NOT NULL	Patents submitted on FDA Form 3542 and listed after August 18, 2003 may have a drug product flag set to 1, indicating the sponsor submitted the patent as claiming the drug product
FK,UK	PATENT_USE_CODE	VARCHAR2(10)		Code to designate a use patent that covers the approved indication or use of a drug product
	DELIST_FLAG	NUMBER	NOT NULL	Sponsor has requested patent be delisted if set to 1. This patent has remained listed because, under Section 505(j)(5)(D)(i) of the Act, a first applicant may retain eligibility for 180-day exclusivity based on a paragraph IV certification to this patent for a certain period. Applicants under Section 505(b)(2) are not required to certify to patents where this flag is set to 1
	SUBMISSION_DATE	DATE		The date on which the FDA receives patent information from the new drug application (NDA) holder. Format is Mmm d, yyyy

PRODUCTS:

Table containing information about approved drug products (from the FDA Orange Book), such as trade name, administration route, approval date. Ingredients in each product are linked to the molecule dictionary via the formulations table.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
	DOSAGE_FORM	VARCHAR2(200)		The dosage form of the product (e.g., tablet, capsule etc)
	ROUTE	VARCHAR2(200)		The administration route of the product (e.g., oral, injection etc)
	TRADE_NAME	VARCHAR2(200)		The trade name for the product
	APPROVAL_DATE	DATE		The FDA approval date for the product (not necessarily first approval of the active ingredient)
	AD_TYPE	VARCHAR2(5)		RX = prescription, OTC = over the counter, DISCN = discontinued
	ORAL	NUMBER		Flag to show whether product is orally delivered
	TOPICAL	NUMBER		Flag to show whether product is topically delivered
	PARENTERAL	NUMBER		Flag to show whether product is parenterally delivered
	BLACK_BOX_WARNING	NUMBER		Flag to show whether the product label has a black box warning
	APPLICANT_FULL_NAME	VARCHAR2(200)		Name of the company applying for FDA approval
	INNOVATOR_COMPANY	NUMBER		Flag to show whether the applicant is the innovator of the product
PK	PRODUCT_ID	VARCHAR2(30)	NOT NULL	FDA application number for the product
	NDA_TYPE	VARCHAR2(10)		New Drug Application Type. The type of new drug application approval. New Drug Applications (NDA or innovator) are "N". Abbreviated New Drug Applications (ANDA or generic) are "A".

PROTEIN_CLASS_SYNONYMS:

Table storing synonyms for the protein family classifications (from various sources including MeSH, ConceptWiki and UMLS).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	PROTCLASSSYN_ID	NUMBER	NOT NULL	Primary key.
FK,UK	PROTEIN_CLASS_ID	NUMBER	NOT NULL	Foreign key to the PROTEIN_CLASSIFICATION table. The protein_class to which this synonym applies.
UK	PROTEIN_CLASS_SYNONYM	VARCHAR2(1000)		The synonym for the protein class.
UK	SYN_TYPE	VARCHAR2(20)		The type or origin of the synonym (e.g., ChEMBL, Concept Wiki, UMLS).

PROTEIN_CLASSIFICATION:

Table storing the protein family classifications for protein targets in ChEMBL (formerly in the target_class table)

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	PROTEIN_CLASS_ID	NUMBER	NOT NULL	Primary key. Unique identifier for each protein family classification.
	PARENT_ID	NUMBER		Protein_class_id for the parent of this protein family.
	PREF_NAME	VARCHAR2(500)		Preferred/full name for this protein family.
	SHORT_NAME	VARCHAR2(50)		Short/abbreviated name for this protein family (not necessarily unique).
	PROTEIN_CLASS_DESC	VARCHAR2(410)	NOT NULL	Concatenated description of each classification for searching purposes etc.
	DEFINITION	VARCHAR2(4000)		Definition of the protein family.
	CLASS_LEVEL	NUMBER	NOT NULL	Level of the class within the hierarchy (level 1 = top level classification)

RELATIONSHIP_TYPE:

Lookup table for assays.relationship_type column, showing whether assays are mapped to targets of the correct identity and species ('Direct') or close homologues ('Homologue')

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	RELATIONSHIP_TYPE	VARCHAR2(1)	NOT NULL	Relationship_type flag used in the assays table
	RELATIONSHIP_DESC	VARCHAR2(250)		Description of relationship_type flags

SITE_COMPONENTS:

Table defining the location of the binding sites in the binding_sites table. A binding site could be defined in terms of which protein subunits (components) are involved, the domains within those subunits to which the compound binds, and possibly even the precise residues involve For a target where the binding site is at the interface of two protein subunits or two domains, there will be two site_components describing each of these subunits/domains.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	SITECOMP_ID	NUMBER	NOT NULL	Primary key.
FK,UK	SITE_ID	NUMBER	NOT NULL	Foreign key to binding_sites table.
FK,UK	COMPONENT_ID	NUMBER		Foreign key to the component_sequences table, indicating which molecular component of the target is involved in the binding site.
FK,UK	DOMAIN_ID	NUMBER		Foreign key to the domains table, indicating which domain of the given molecular component is involved in the binding site (where not known, the domain_id may be null).
	SITE_RESIDUES	VARCHAR2(2000)		List of residues from the given molecular component that make up the binding site (where not know, will be null).

SOURCE:

Table showing source from which ChEMBL data is derived (e.g., literature, deposited datasets, patents, drug or clinical candidate sources etc)

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	SRC_ID	NUMBER	NOT NULL	Identifier for each source (used in compound_records and assays tables)
	SRC_DESCRIPTION	VARCHAR2(500)		Descriptive name for the source
	SRC_SHORT_NAME	VARCHAR2(20)		A short name for each data source, for display purposes
	SRC_COMMENT	VARCHAR2(1200)		Additional notes on the source
	SRC_URL	VARCHAR2(200)		URL(s) pointing to website(s) with relevant information about the source
	DDID_PATTERN	VARCHAR2(2000)		Pattern used to construct RIDX and AIDX. E.g. TARGET_DATE_AUTHOR

STRUCTURAL_ALERT_SETS:

Table showing list of sets of structural alerts that have been included in COMPOUND_STRUCTURAL_ALERT table. It should be noted some alerts/alert sets are more permissive than others and may flag a large number of compounds. Results should be interpreted with care, depending on the use-case, and not treated as a blanket filter (e.g., around 50% of approved drugs have 1 or more alerts from these sets).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ALERT_SET_ID	NUMBER	NOT NULL	Unique ID for the structural alert set
UK	SET_NAME	VARCHAR2(100)	NOT NULL	Name (or origin) of the structural alert set
	PRIORITY	NUMBER	NOT NULL	Priority assigned to the structural alert set for display on the ChEMBL interface (priorities >=4 are shown by default).

STRUCTURAL_ALERTS:

Table storing a list of structural features (encoded as SMARTS) that are potentially undesirable in drug discovery context. It should be noted some alerts/alert sets are more permissive than others and may flag a large number of compounds. Results should be interpreted with care, depending on the use-case, and not treated as a blanket filter (e.g., around 50% of approved drugs have 1 or more alerts from these sets).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	ALERT_ID	NUMBER	NOT NULL	Primary key. Unique identifier for the structural alert
FK,UK	ALERT_SET_ID	NUMBER	NOT NULL	Foreign key to structural_alert_sets table indicating which set this particular alert comes from
UK	ALERT_NAME	VARCHAR2(100)	NOT NULL	A name for the structural alert
UK	SMARTS	VARCHAR2(4000)	NOT NULL	SMARTS defining the structural feature that is considered to be an alert

TARGET_COMPONENTS:

Links molecular target from the target_dictionary to the components they consist of (in the component_sequences table). For a protein complex or protein family target, for example, there will be multiple protein components in the component_sequences table.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
FK,UK	TID	NUMBER	NOT NULL	Foreign key to the target_dictionary, indicating the target to which the components belong.
FK,UK	COMPONENT_ID	NUMBER	NOT NULL	Foreign key to the component_sequences table, indicating which components belong to the target.
PK	TARGCOMP_ID	NUMBER	NOT NULL	Primary key.
	HOMOLOGUE	NUMBER	NOT NULL	Indicates that the given component is a homologue of the correct component (e.g., from a different species) when set to 1. This may be the case if the sequence for the correct protein/nucleic acid cannot be found in sequence databases. A value of 2 indicates that the sequence given is a representative of a species group, e.g., an E. coli protein to represent the target of a broad-spectrum antibiotic.

TARGET_DICTIONARY:

Target Dictionary containing all curated targets for ChEMBL. Includes both protein targets and non-protein targets (e.g., organisms, tissues, cell lines)

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	TID	NUMBER	NOT NULL	Unique ID for the target
FK	TARGET_TYPE	VARCHAR2(30)		Describes whether target is a protein, an organism, a tissue etc. Foreign key to TARGET_TYPE table.
	PREF_NAME	VARCHAR2(200)	NOT NULL	Preferred target name: manually curated
	TAX_ID	NUMBER		NCBI taxonomy id of target
	ORGANISM	VARCHAR2(150)		Source organism of molecuar target or tissue, or the target organism if compound activity is reported in an organism rather than a protein or tissue
FK,UK	CHEMBL_ID	VARCHAR2(20)	NOT NULL	ChEMBL identifier for this target (for use on web interface etc)
	SPECIES_GROUP_FLAG	NUMBER	NOT NULL	Flag to indicate whether the target represents a group of species, rather than an individual species (e.g., 'Bacterial DHFR'). Where set to 1, indicates that any associated target components will be a representative rather than a comprehensive set.

TARGET_RELATIONS:

Table showing relationships between different protein targets based on overlapping protein components (e.g., relationship between a protein complex and the individual subunits).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
FK	TID	NUMBER	NOT NULL	Identifier for target of interest (foreign key to target_dictionary table)
	RELATIONSHIP	VARCHAR2(20)	NOT NULL	Relationship between two targets (e.g., SUBSET OF, SUPERSET OF, OVERLAPS WITH)
FK	RELATED_TID	NUMBER	NOT NULL	Identifier for the target that is related to the target of interest (foreign key to target_dicitionary table)
PK	TARGREL_ID	NUMBER	NOT NULL	Primary key

TARGET_TYPE:

Lookup table for target types used in the target dictionary

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	TARGET_TYPE	VARCHAR2(30)	NOT NULL	Target type (as used in target dictionary)
	TARGET_DESC	VARCHAR2(250)		Description of target type
	PARENT_TYPE	VARCHAR2(25)		Higher level classification of target_type, allowing grouping of e.g., all 'PROTEIN' targets, all 'NON-MOLECULAR' targets etc.

TISSUE_DICTIONARY:

Table storing information about tissues used in assays.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	TISSUE_ID	NUMBER	NOT NULL	Primary key, numeric ID for each tissue.
UK	UBERON_ID	VARCHAR2(15)		Uberon ontology identifier for this tissue.
UK	PREF_NAME	VARCHAR2(200)	NOT NULL	Name for the tissue (in most cases Uberon name).
UK	EFO_ID	VARCHAR2(20)		Experimental Factor Ontology identifier for the tissue.
FK,UK	CHEMBL_ID	VARCHAR2(20)	NOT NULL	ChEMBL identifier for this tissue (for use on web interface etc)
	BTO_ID	VARCHAR2(20)		BRENDA Tissue Ontology identifier for the tissue.
	CALOHA_ID	VARCHAR2(7)		Swiss Institute for Bioinformatics CALOHA Ontology identifier for the tissue.

USAN_STEMS:

Table storing definitions for stems used in USANs (United States Adopted Names).

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	USAN_STEM_ID	NUMBER	NOT NULL	Numeric primary key.
UK	STEM	VARCHAR2(100)	NOT NULL	Stem defined for use in United States Adopted Names.
UK	SUBGROUP	VARCHAR2(100)		More specific subgroup of the stem defined for use in United States Adopted Names.
	ANNOTATION	VARCHAR2(2000)		Meaning of the stem (e.g., the class of compound it applies to).
	STEM_CLASS	VARCHAR2(100)		Indicates whether stem is used as a prefix/infix/suffix/combined prefix and suffix

VARIANT_SEQUENCES:

Table storing information about mutant sequences and other variants used in assays. The sequence provided is a representative sequence incorporating the reported mutation/variant used in the assay - it is not necessarily the exact sequence used in the experiment.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	VARIANT_ID	NUMBER	NOT NULL	Primary key, numeric ID for each sequence variant; -1 for unclassified variants.
UK	MUTATION	VARCHAR2(2000)		Details of variant(s) used, with residue positions adjusted to match provided sequence.
UK	ACCESSION	VARCHAR2(25)		UniProt accesion for the representative sequence used as the base sequence (without variation).
	VERSION	NUMBER		Version of the UniProt sequence used as the base sequence.
	ISOFORM	NUMBER		Details of the UniProt isoform used as the base sequence where relevant.
	SEQUENCE	CLOB		Variant sequence formed by adjusting the UniProt base sequence with the specified mutations/variations.
	ORGANISM	VARCHAR2(200)		Organism from which the sequence was obtained.
	TAX_ID	NUMBER		NCBI Tax ID for the organism from which the sequence was obtained

VERSION:

Table showing release version and creation date for the database, and associated ontologies and packages.

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	NAME	VARCHAR2(50)	NOT NULL	Name of release version
	CREATION_DATE	DATE		Date database created
	COMMENTS	VARCHAR2(2000)		Description of release version

WARNING_REFS:

Table storing references indicating the source of drug warning information

KEYS	COLUMN_NAME	DATA_TYPE	NULLABLE	COMMENT
PK	WARNREF_ID	NUMBER	NOT NULL	Primary key for the warning reference
FK	WARNING_ID	NUMBER		Foreign key to the drug_warning table
	REF_TYPE	VARCHAR2(50)		Type/source of reference
	REF_ID	VARCHAR2(4000)		Identifier for the reference in the source
	REF_URL	VARCHAR2(4000)		Full URL linking to the reference